

Giacomo Cappelletto

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EDUCATION

Boston University

B.S. in Computer Engineering; GPA: 3.97/4.00

Expected May 2028

Boston, MA

RESEARCH & EXPERIENCE

Boston University - Prof. Brian Kulis

Undergraduate Machine Learning Researcher

Aug 2026 – Present

Boston, MA

- Researching robust contextual metric learning for human-motion retrieval, learning fixed-length embeddings from temporal 2D/3D pose sequences for nearest-neighbor search across labeled movement classes.
- Comparing contextual and contrastive objectives with triplet, supervised-contrastive, and classification baselines; evaluating Recall@K and mAP under pose noise, missing joints, temporal jitter, occlusion, and viewpoint shift.

Boston University College of Engineering

Undergraduate Machine Learning Researcher

Jan 2026 – May 2026

Boston, MA

- Built a Python vision pipeline to stabilize rowing video, run single-athlete 2D keypoint inference with MMPose, lift poses to 3D skeletons with MotionBERT, and output per-frame joint positions and angles.
- Computed stroke-averaged 3D kinematics and trained sequence-to-sequence models mapping time-aligned motion to force curves, validated against instrumented on-water and erg telemetry.

Banca Mediolanum

Software & AI Engineering Intern, Customer Knowledge

Jun 2026 – Aug 2026

Milan, Italy

- Built and deployed a LangGraph/MLflow agent with governed tools, retrieval, typed model interfaces, evaluation, evidence review, and end-to-end tracing on Databricks Model Serving.
- Developed reproducible evaluation workflows with versioned prompts, side-by-side benchmarks, LLM judging, MLflow GenAI scoring, and Unity Catalog quality controls.

Societa Cappelletto S.r.l.

Software Engineer

May 2025 – Oct 2025

Treviso, Italy

- Rebuilt a production inventory application in Tauri, React, and Rust, reducing application size by 70%, memory use by 80%, startup time by 60%, and search latency by 43%.

SELECTED PROJECTS

Deskinator - Autonomous Desk-Cleaning Robot

Python, Raspberry Pi

- Built autonomy software with APDS9960 edge sensing, stepper odometry, RANSAC rectangle fitting, boustrophedon coverage planning, telemetry, and simulation-based performance analysis.

CIFAR-10 Neural Architecture Search

TensorFlow/Keras

- Built a pipeline to generate, train, evaluate, and select CNN candidates with automated metric logging and best-model selection.

LEADERSHIP & ATHLETICS

Boston University Men's Rowing | NCAA Division I Student-Athlete | Sep 2024 – Present

- Freshman Student-Athlete of the Year (2025); Most Improved Oarsman (2026).

SKILLS

Languages: Python, C/C++, Rust, SQL, TypeScript/JavaScript

ML/CV: PyTorch, TensorFlow/Keras, MMPose, MotionBERT, metric learning, contrastive learning, pose estimation, 3D motion, sequence modeling, RANSAC

Platforms: MLflow, Databricks, LangGraph, PostgreSQL, Docker, Google Cloud Run, Git